CCT College Dublin

Assessment Cover Page

Module Title:	Strategic Business Information Technology Problem Solving for Industry (Capstone Project)
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Lecturer Name:	Ken Healy Muhammad Iqbal
Students Full Name:	Sander Luiz Santos Soares Thiago Gonçalves da Costa
Students Number:	2022164 2022161
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Declaration

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1. Project Concept

The proposed project aims to develop a *Lightweight Network Intrusion Detection System (IDS)* that uses machine learning techniques to detect suspicious or potentially malicious network traffic. It will be able to detect some Cybersecurity threats such as DoS (denial-of-service attacks), malware distribution, and web attacks, as these attacks are becoming more common, and many organizations have difficulties detecting these threats quickly using traditional monitoring tools.

Ireland is facing an increasing problem, according to the recent assessment provided by the National Cyber Security Centre (NCSC) which states that there is a growing risk of cyber-attacks across Ireland for two reasons, firstly the increased digitalisation of services and secondly because of the increasing sophistication of cyber-crime. The report states that attacks on critical sectors including healthcare, energy, transport and financial services could disrupt essential services and create a loss of trust in institutions providing these services (O'Donovan, 2025).

The project will analyse network traffic data to identify abnormal patterns that may indicate cyber-attacks. The machine learning model will be trained by using an existing cyber security dataset to distinguish between normal network behaviour and malicious activity. The system will then assign a risk or anomaly score to network flows and identify potentially dangerous connections, such as suspicious IP addresses or unusual port activity.

The final product would be a prototype monitoring dashboard that displays security alerts, suspicious network behaviour, and important traffic statistics. Being a lightweight tool, we can target some specific customers, including small and medium-sized enterprises (SMEs), schools, clinics, small offices, and managed service providers that often have limited security staff and restricted cybersecurity budgets.

2. Business Analysis (SWOT Analysis)

To analyse the potential commercial values of our system on the market, a SWOT analysis is used. SWOT is a strategic business analysis tool that evaluates the Strengths, Weaknesses, Opportunities, and Threats of a product or service. This method helps to understand how the system could create value for organisations and what advantages it may provide in the cybersecurity market (Kenton, 2025).

2.1 Strengths

One of the main strengths is its ability to automatically detect suspicious network behaviour using machine learning techniques. Security teams of many organizations struggle monitoring all network traffic manually, especially because even small companies generate large amounts of network traffic and having this process automated can support faster detection of cyber threats, improving the operational efficiency and reducing risk of cyber-attacks.

Another, and maybe our main strength, is that the system is designed to be lightweight and relatively easy to deploy, which makes it attractive for organizations that do not have large cybersecurity teams or expensive security infrastructure, making it an accessible alternative for small and medium-sized businesses.

2.2 Weaknesses

Machine learning uses historical data, in other words, the model is trained using a dataset and cannot update itself automatically. To remain effective, it would require continuous retraining with new data. Since cyber threats evolve quickly, the system may not detect new attack techniques if the model is not updated regularly.

2.3 Opportunities

Cybersecurity threats are increasing globally, creating strong demand for solutions that improve threat detection. In Ireland, the National Cyber Security Centre has warned about the increasing risk of cyber-attacks targeting critical sectors such as healthcare, energy, and financial services (O'Donovan, 2025).

This creates opportunities for systems that can improve visibility of network activity and detect attacks faster, especially for small to medium-sized organizations that do not have advanced cybersecurity infrastructure. Such tool can significantly reduce operational workload and improve response time to security incidents.

2.4 Threats

Some large cybersecurity companies already offer intrusion detection and monitoring platforms, which can be a potential threat.

Also, organizations may prefer integrated security platforms rather than standalone tools. However, being a lightweight and cost-effective system could still attract organisations that need affordable cybersecurity solutions.

3. Technology Selection and Justification

The chosen key technologies for this project are Python with Scikit-learn and Streamlit. This combination fits the purpose of creating a lightweight Network Intrusion Detection System (IDS) that can identify suspicious network activity and provide the results in a simple dashboard. Choosing technologies that are effective, affordable, and easy to implement is important because technology adoption in SMEs is strongly influenced by cost, resources, infrastructure, and business value (Zamani, 2022).

For data preprocessing, model training, testing and traffic classification, Python with Scikit-learn was selected. Scikit-learn is an open-source machine learning library that supports supervised and unsupervised learning, preprocessing, model selection and evaluation, making it suitable for structured network traffic data (Scikit-learn, 2025). It has lower computational overhead and is easier to implement than more advanced machine learning platforms. It is also well suited to structured flow-based data, which makes it a strong choice for this IDS prototype.

The main alternative considered for this technology was MATLAB, which is a strong platform for technical computing and data analysis, but it is proprietary software and commercial use may involve licence costs, which can be a problem for small businesses or organisations with limited budgets that may struggle to afford the software (MathWorks, 2019). On the other hand, Python and Scikit-learn are both open-source, meaning that they are more accessible, more cost-effective, and reduce dependence on a single vendor.

Streamlit was chosen to build the dashboard interface of the system, which includes suspicious IP addresses, unusual ports, traffic summaries and an alerts list with a CSV export.

Streamlit is an open-source Python framework that enables data apps to be created quickly, making it suitable for presenting the results of machine learning algorithms in a simple and practical way (Streamlit, 2024). This also helps present the machine learning outputs in a usable interface for IT staff who may not have much knowledge of machine learning.

The main alternative considered for this technology was Power BI, which is widely used for dashboards and reporting in business environments, but it is less appropriate for a tightly integrated machine learning prototype. Microsoft has different paid plans such as Power BI Pro and Power BI Premium Per User, which bring cost considerations for smaller organisations (Microsoft, 2025). In addition, Streamlit integrates more directly with the Python-based machine learning workflow, making it easier to link the model output to the dashboard in one environment.

Overall, these technologies were selected because they support both the technical and business aims of the project. Compared with MATLAB and Power BI, the chosen technologies are more affordable, more flexible, and more appropriate for a lightweight prototype for SMEs, with a lower barrier to adoption for resource-constrained organisations.

4. Legal and Ethical Issues

There are important legal and ethical issues that need to be considered when designing and building this system. Although a public dataset will be used during the initial stages to train and test the system, in a real-world implementation of a lightweight Network Intrusion Detection System (IDS) the system could be used to monitor live network traffic. This means the system may process information such as IP addresses, device identifiers, timestamps and patterns of network usage. In some cases, this information may be linked to specific people, which means that privacy and data protection issues can occur under GDPR principles such as lawful processing, data minimisation, secure storage, and controlled access (GDPR, 2018).

For this reason, the system should follow a privacy by design approach (GDPR, 2018b). An important point is that the project is based on network flow data and metadata, rather than deep inspection of the entire content of communications. This helps reduce the amount of personal or sensitive information being processed. In practice, the system should focus only on the data needed to detect suspicious behaviour instead of collecting unnecessary information (ICO, 2024). If used in a real environment, access to logs and alerts should be restricted and data should only be retained for a limited time. Monitoring practices should be transparent if used in a real-world scenario, and organizations should clearly specify who can access logs and alerts. To reduce privacy risks, anonymisation or pseudonymisation should also be considered.

There are also ethical issues associated with the use of machine learning in intrusion detection. One concern is the risk of false positives, where legitimate traffic could be incorrectly flagged as suspicious. This might disrupt normal business activity, cause confusion for users or add extra workload to IT staff. On the other hand, false negatives could allow real malicious activity to go unnoticed, creating a false sense of security (NIST, 2024). For this reason, the system should be used as a decision-support tool, not a fully autonomous system. Human administrators should make the final decision on whether to block traffic or take action, especially in smaller organisations where errors could affect essential services.

Another ethical issue is model bias and lack of generalisation. The proposed system is trained using the CIC-IDS2017 dataset, which is very useful for research and prototyping, but is still based on simulated network behaviour. This means that the model may not reflect the actual infrastructure, user behaviour or threat patterns of each organisation. In addition, the dataset was developed in 2017 and may not accurately reflect newer attack methods and current cyber threats. Because of this, the model may be good in testing but not so good in actual deployment (Engelen et al., 2021). It is important to be clear about these limitations and not present the system as a complete or fully accurate security solution.

Overall, these issues do not prevent the system from being developed, but they show that it must be designed and used responsibly, especially if it is used by SMEs or other organisations with limited cybersecurity resources.

5. Data Collection & Competitive Advantage

The dataset is obtained from a public cybersecurity dataset called CIC-IDS2017, developed by the Canadian Institute for Cybersecurity. It contains realistic network traffic data including both normal activity and several types of cyber-attacks such as brute force, DoS, DDoS, botnet, infiltration and web attacks (Sharafaldin, et al., 2018).

The data presented in this dataset is a simulation of realistic network behaviour. Researchers generated the network traffic by modelling the behaviour of 25 users' interaction with common internet protocols, such as HTTP, HTTPS, FTP, SSH, and email services. The data was collected over the weekdays (Monday to Friday) from July 3rd to July 7th, when the first day had only normal activity while the remaining had different types of cyber-attacks executed at specific times (Sharafaldin, et al., 2018).

Being created on a simulated environment can be one of the limitations of our dataset, so it may not fully represent the real network conditions of every organisation. However, Organisations can gain a better understanding of their internal network behaviour by analysing this network traffic data. The approach can assist prioritise unusual security resources more efficiently, identify high-risk hosts or services earlier, and spot persistent suspicious activity over time. By highlighting unusual traffic patterns, it can help speed up security decisions and cut down on the amount of time spent manually studying records. Having the most sophisticated cybersecurity infrastructure may not give small and medium-sized businesses a competitive edge; instead, they may benefit from being able to identify issues early and respond with fewer resources. Increased visibility of dubious traffic can boost customer and stakeholder trust, lessen operational disturbance, and promote service reliability. Additionally, the technology does not only support security, but also improves operational efficiency, resilience, and affordability, which are important factor for resource-controlled organisations.

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